THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 90°	One part $= 18^\circ$
2	**(a)** 60°	$x + 2x = 180$
3	**(c)** 130°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
4	**(b)** 50°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
5	**(c)** 110°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$
6	**(b)** 6	Perimeter $= 4\times$ side
7	**(d)** isosceles trapezium	Isosceles trapezium has equal diagonals
8	**(d)** 120°	One part $= 30^\circ$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Co-interior angles between parallel lines add up to 180°, which gives A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but bisecting does not give equal lengths; equality follows from the right angles of the rectangle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite angles equal gives $x = 12$ (1); $\angle A = 62^\circ$, so $\angle B = 180 - 62 = 118^\circ$ (1).	1+1
12	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 126^\circ$, which is obtuse (1).	1+1
13	$\angle D = \angle B = 100^\circ$ (1); $\angle C = 360 - 70 - 2 \times 100 = 90^\circ$ (1).	1+1
14	Opposite angles equal gives $x = 10$ (1); $\angle A = 60^\circ$, so $\angle B = 180 - 60 = 120^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 130^\circ$ (1); $\angle PAB = 25^\circ$ and $\angle PBA = 65^\circ$ (1); $\angle APB = 180 - (25 + 65) = 90^\circ$ (1).	1+1+1
16	$10x - 60 = 360$ (1); $x = 42$ (1); angles 42°, 74°, 106°, 138° (1).	1+1+1
17	Let the smaller be x: $x + 2x = 180$ (1); $x = 60^\circ$ (1); angles 60°, 120°, 60°, 120° (1).	1+1+1
	OR: $\angle B = 110^\circ$ (1); $\angle PAB = 35^\circ$ and $\angle PBA = 55^\circ$ (1); $\angle APB = 180 - (35 + 55) = 90^\circ$ (1).	1+1+1
18	$10x - 120 = 360$ (1); $x = 48$ (1); angles 48°, 76°, 104°, 132° (1).	1+1+1
	OR: Let the smaller be x: $x + 3x = 180$ (1); $x = 45^\circ$ (1); angles 45°, 135°, 45°, 135° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Opposite angles are equal, $\angle C = 70^\circ$ (1). (ii) $\angle B = 180^\circ - 70^\circ = 110^\circ$ (1). (iii) $2(80 + 50) = 260$ cm (1). (iv) Diagonals of a parallelogram bisect each other, so $OA = 48$ cm and $OB = 30$ cm (1+1).	1+1+1+2
20	One part $= 360 \div 12 = 30^\circ$ (1); angles 60°, 90°, 90°, 120° (2); $\angle A + \angle D = 60 + 120 = 180^\circ$ (1); these are co-interior angles, so AB $\parallel$ DC (1).	1+2+1+1
	OR: Opposite sides are equal: $4x + 2 = 3x + 9$ gives $x = 7$ (2); AB = CD $= 30$ cm (1); perimeter $= 2(30 + 15) = 90$ cm (2).	2+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 55°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
2	**(c)** 116°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$
3	**(b)** 4	Perimeter $= 4\times$ side
4	**(b)** The diagonals are equal	Rhombus diagonals are equal only when it is a square
5	**(b)** 60°	One part $= 15^\circ$
6	**(a)** 36°	$x + 4x = 180$
7	**(b)** 104°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
8	**(c)** 60°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
9	**(c)** A is true but R is false.	A is true (all angles become 90°). R is false: equal diagonals make a parallelogram a rectangle, not a rhombus.
10	**(d)** A is false but R is true.	A is false (equal diagonals belong to a rectangle or square); R is a true property of the rhombus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 144^\circ$, which is obtuse (1).	1+1
12	$\angle D = \angle B = 96^\circ$ (1); $\angle C = 360 - 84 - 2 \times 96 = 84^\circ$ (1).	1+1
13	Opposite angles equal gives $x = 5$ (1); $\angle A = 33^\circ$, so $\angle B = 180 - 33 = 147^\circ$ (1).	1+1
14	One part $= 360 \div 24 = 15^\circ$ (1); largest $= 120^\circ$, which is obtuse (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 96^\circ$ (1); $\angle PAB = 42^\circ$ and $\angle PBA = 48^\circ$ (1); $\angle APB = 180 - (42 + 48) = 90^\circ$ (1).	1+1+1
16	$10x - 30 = 360$ (1); $x = 39$ (1); angles 39°, 73°, 107°, 141° (1).	1+1+1
17	Let the smaller be x: $x + 4x = 180$ (1); $x = 36^\circ$ (1); angles 36°, 144°, 36°, 144° (1).	1+1+1
	OR: $\angle B = 84^\circ$ (1); $\angle PAB = 48^\circ$ and $\angle PBA = 42^\circ$ (1); $\angle APB = 180 - (48 + 42) = 90^\circ$ (1).	1+1+1
18	$10x - 90 = 360$ (1); $x = 45$ (1); angles 45°, 75°, 105°, 135° (1).	1+1+1
	OR: Let the smaller be x: $x + 5x = 180$ (1); $x = 30^\circ$ (1); angles 30°, 150°, 30°, 150° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 90°, as diagonals of a rhombus are perpendicular (1). (ii) Half-diagonals 8 cm and 6 cm, side $= \sqrt{8^2 + 6^2} = 10$ cm (1). (iii) $4 \times 10 = 40$ cm (1). (iv) Opposite angle 64°, adjacent angles 116° each (1); $64 + 116 + 64 + 116 = 360$ (1).	1+1+1+2
20	One part $= 360 \div 20 = 18^\circ$ (1); angles 72°, 90°, 90°, 108° (2); $\angle A + \angle D = 72 + 108 = 180^\circ$ (1); these are co-interior angles, so AB $\parallel$ DC (1).	1+2+1+1
	OR: Opposite sides are equal: $6x + 1 = 4x + 11$ gives $x = 5$ (2); AB = CD $= 31$ cm (1); perimeter $= 2(31 + 14) = 90$ cm (2).	2+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** square	Both rhombus and rectangle properties
2	**(b)** 100°	One part $= 20^\circ$
3	**(c)** 30°	$x + 5x = 180$
4	**(b)** 98°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
5	**(c)** 65°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
6	**(c)** 118°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$
7	**(a)** 3	Perimeter $= 4\times$ side
8	**(a)** rhombus	All sides equal
9	**(a)** Both A and R are true and R is the correct explanation of A.	$1 + 2 + 3 + 4 = 10$ parts, one part $= 36^\circ$, largest $= 4 \times 36^\circ = 144^\circ$; R is used.
10	**(c)** A is true but R is false.	A is true; R is false, as only the opposite angles of a rhombus are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle D = \angle B = 115^\circ$ (1); $\angle C = 360 - 55 - 2 \times 115 = 75^\circ$ (1).	1+1
12	Opposite angles equal gives $x = 20$ (1); $\angle A = 95^\circ$, so $\angle B = 180 - 95 = 85^\circ$ (1).	1+1
13	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 162^\circ$, which is obtuse (1).	1+1
14	$\angle D = \angle B = 80^\circ$ (1); $\angle C = 360 - 100 - 2 \times 80 = 100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 70^\circ$ (1); $\angle PAB = 55^\circ$ and $\angle PBA = 35^\circ$ (1); $\angle APB = 180 - (55 + 35) = 90^\circ$ (1).	1+1+1
16	$10x - 150 = 360$ (1); $x = 51$ (1); angles 51°, 77°, 103°, 129° (1).	1+1+1
17	Let the smaller be x: $x + 8x = 180$ (1); $x = 20^\circ$ (1); angles 20°, 160°, 20°, 160° (1).	1+1+1
	OR: $\angle B = 56^\circ$ (1); $\angle PAB = 62^\circ$ and $\angle PBA = 28^\circ$ (1); $\angle APB = 180 - (62 + 28) = 90^\circ$ (1).	1+1+1
18	$10x - 180 = 360$ (1); $x = 54$ (1); angles 54°, 78°, 102°, 126° (1).	1+1+1
	OR: Let the smaller be x: $x + 11x = 180$ (1); $x = 15^\circ$ (1); angles 15°, 165°, 15°, 165° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 360° (1). (ii) $12x = 360^\circ$, $x = 30^\circ$ (1). (iii) $4x = 120^\circ$ (1). (iv) Angles are $60^\circ, 90^\circ, 90^\circ, 120^\circ$; opposite angles A and C are not equal, so it is not a parallelogram (1). $\angle A + \angle D = 180^\circ$ so $AB \parallel DC$ (it is a trapezium) (1).	1+1+1+2
20	One part $= 360 \div 30 = 12^\circ$ (1); angles 72°, 84°, 96°, 108° (2); $\angle A + \angle D = 72 + 108 = 180^\circ$ (1); these are co-interior angles, so AB $\parallel$ DC (1).	1+2+1+1
	OR: Opposite sides are equal: $5x + 2 = 2x + 17$ gives $x = 5$ (2); AB = CD $= 27$ cm (1); perimeter $= 2(27 + 16) = 86$ cm (2).	2+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

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